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Final Project Presentation

198888 PR Data Analysis Lab

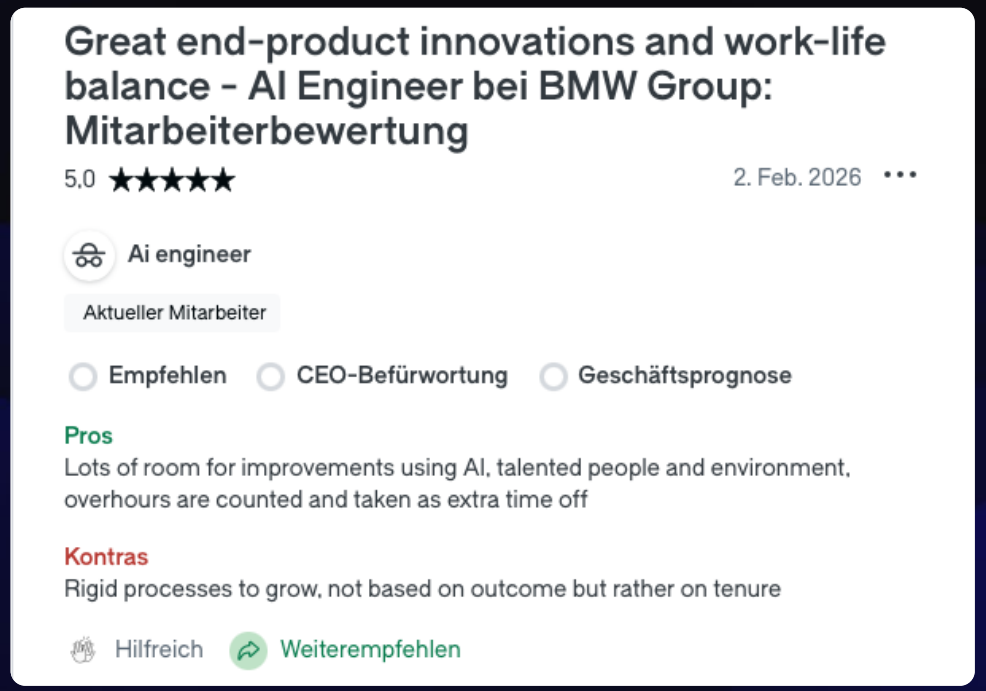
Franziska Zuber

Innsbruck, 26.06.2026

Identifying the drivers of employee satisfaction in text: a comparative analysis of an LLM and a dictionary-based approach on kununu and glassdoor reviews



Example Reviews



Great end-product innovations and work-life balance - AI Engineer bei BMW Group: Mitarbeiterbewertung

5,0 ★★★★★ 2. Feb. 2026 •••

Ai engineer

Aktueller Mitarbeiter

Empfehlen CEO-Befürwortung Geschäftsprognose

Pros

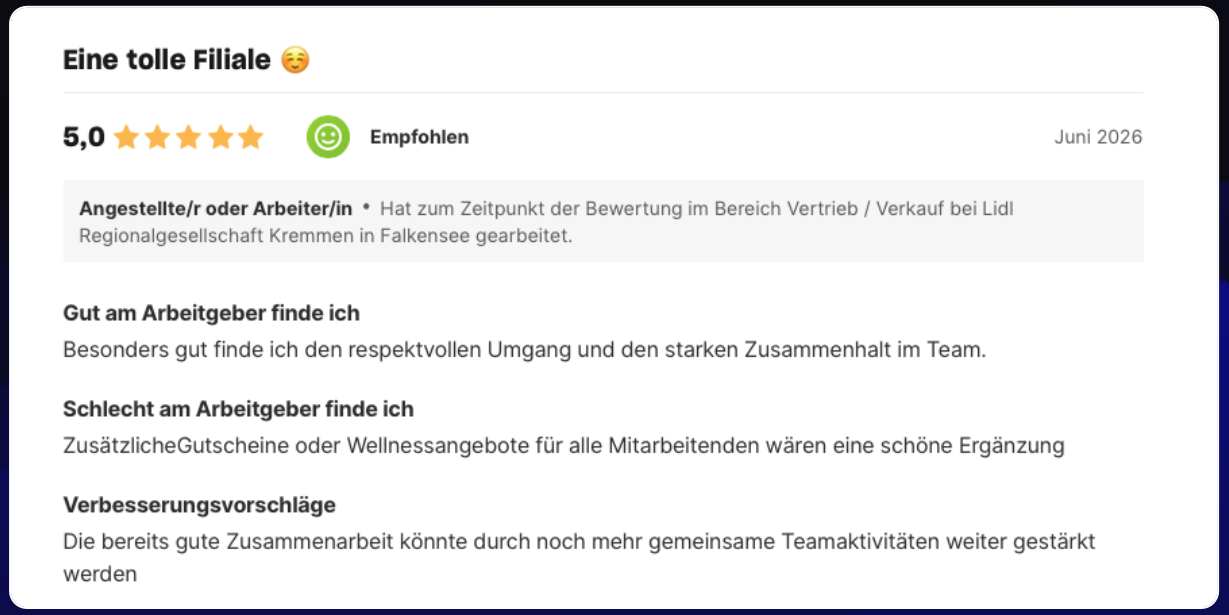
Lots of room for improvements using AI, talented people and environment, overhours are counted and taken as extra time off

Kontras

Rigid processes to grow, not based on outcome but rather on tenure

Hilfreich Weiterempfehlen

GLASSDOOR



Eine tolle Filiale 😌

5,0 ★★★★★ Empfohlen Juni 2026

Angestellte/r oder Arbeiter/in • Hat zum Zeitpunkt der Bewertung im Bereich Vertrieb / Verkauf bei Lidl Regionalgesellschaft Kremmen in Falkensee gearbeitet.

Gut am Arbeitgeber finde ich

Besonders gut finde ich den respektvollen Umgang und den starken Zusammenhalt im Team.

Schlecht am Arbeitgeber finde ich

ZusätzlicheGutscheine oder Wellnessangebote für alle Mitarbeitenden wären eine schöne Ergänzung

Verbesserungsvorschläge

Die bereits gute Zusammenarbeit könnte durch noch mehr gemeinsame Teamaktivitäten weiter gestärkt werden

(just an excerpt)

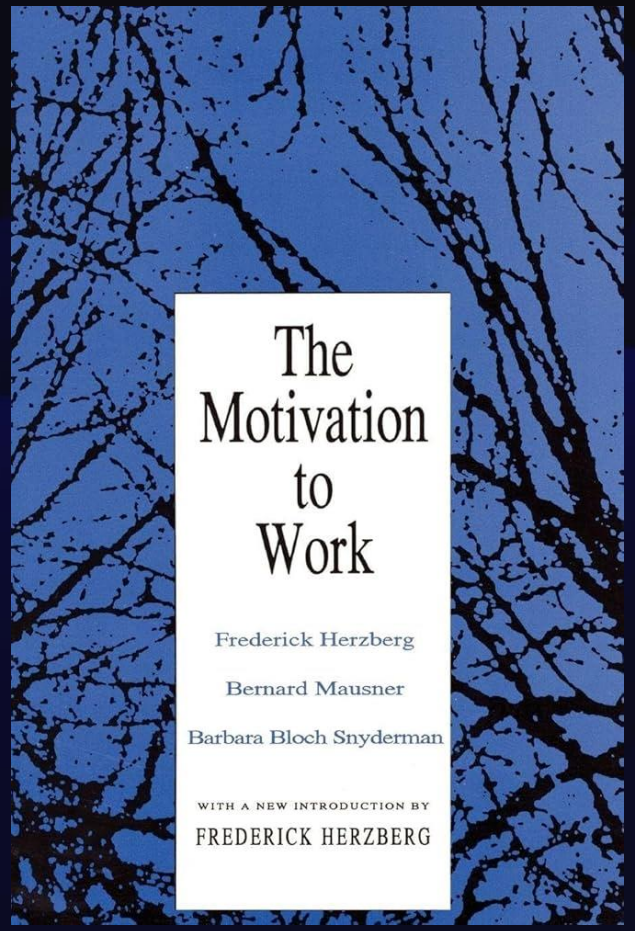
kununu

Example Reviews


```
  {
    "rating": "1.0",
    "pros": "None. We just keep losing.",
    "cons": "Your health insurance may very well lapse repeatedly if your company is acquired by Seimens. The company I work for
has been without health insurance for over 14 days altogether so far in 2026, with another lapse and our third insurance company
switch expected next month. This has made it very difficult to make appointments, refill prescriptions, and to get urgent medical
care for ourselves and our families. These issues are also causing ongoing disruptions during our operating hours while we are
navigating a big transition. This is unacceptable behavior for any company.",
    "position": "Shipping and receiving",
    "employment_status": "Former employee, less than 1 year"
  },
  {
    "rating": "5.0",
    "pros": "Least toxic environment to work in",
    "cons": "the growth personally will be little slow",
    "position": "Anonymous employee",
    "employment_status": "Current employee, more than 1 year"
  },
```


textual variables of relevance: „pros“ and „cons“

Theoratical backround:
Why is employee-review data relevant?



Motivation Factors:

Achievement
Recognition
Work itself
Responsibility
Growth

↓

employee satisfaction

★★★

Hygiene Factors:

Working conditions
Salary/benefits
Job security
Company policy
Relationships

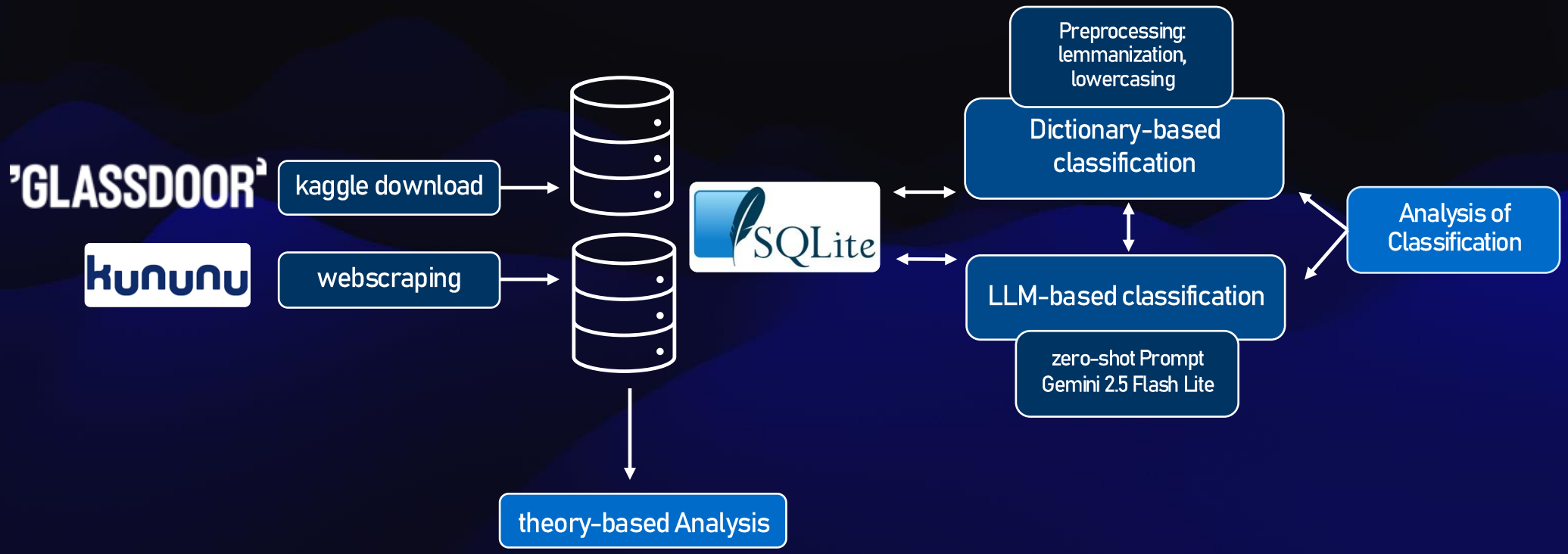
↓

employee dissatisfaction

★☆☆

(Herzberg et al., 1959)

Workflow

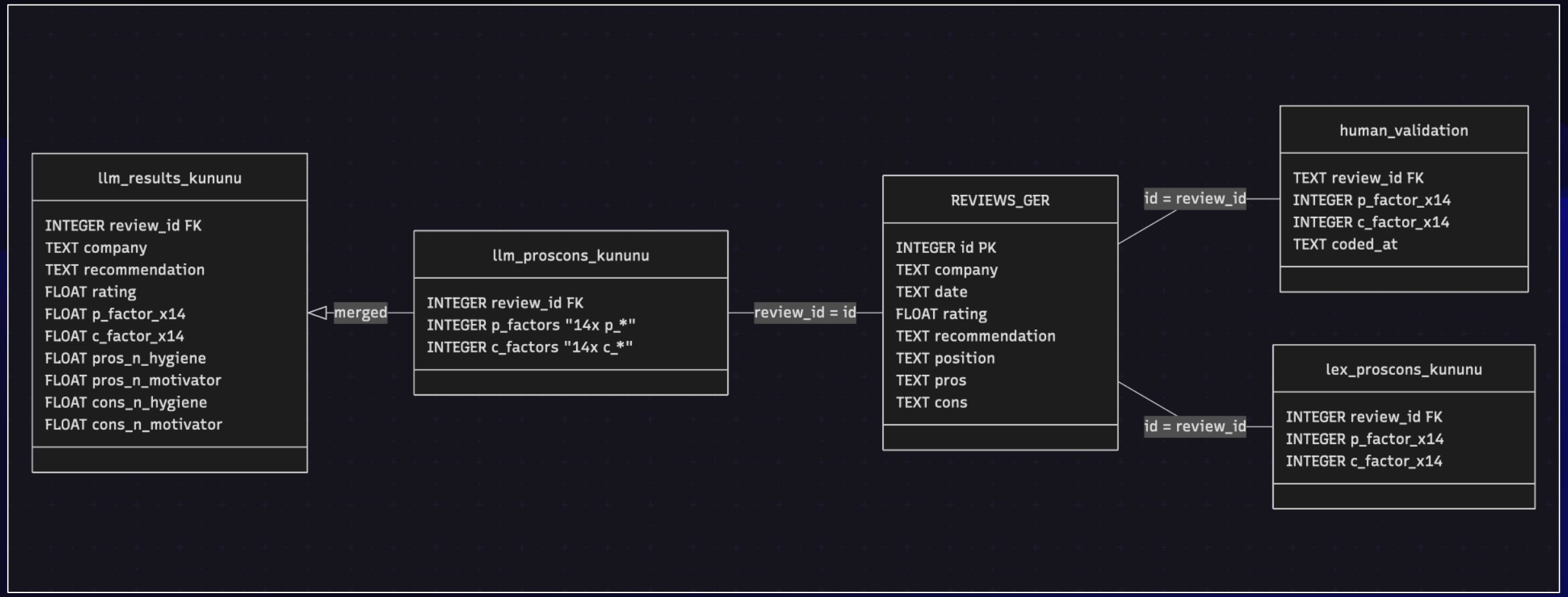

GLASSDOOR
kaggle download
kununu
webscraping
SQLite
Preprocessing: lemmanization, lowercasing
Dictionary-based classification
LLM-based classification
zero-shot Prompt Gemini 2.5 Flash Lite
Analysis of Classification
theory-based Analysis

Classification

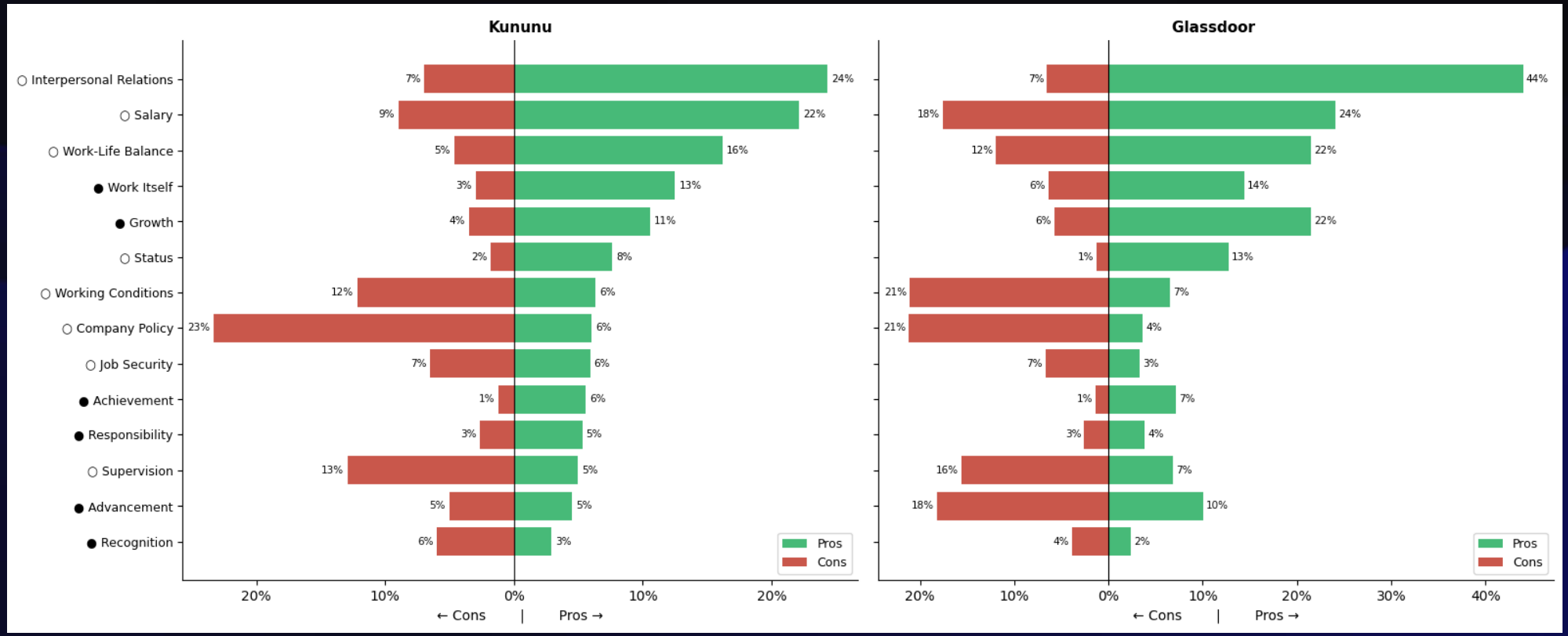
Example for a word-list:

```
"salary": [
    "gehalt", "lohn", "vergütung", "bonus", "benefits", "benefit",
    "bezahlung", "urlaubsgeld", "weihnachtsgeld", "tarifvertrag",
    "gehaltserhöhung", "einkommen", "entlohnung", "provision",
    "betriebsrente", "altersvorsorge", "sonderzahlung", "gehälter",
    "löhne", "gehaltsgefüge", "gehaltsstruktur", "lohnerhöhung",
    "gute bezahlung", "schlechte bezahlung", "faire bezahlung",
    "marktgerechtes gehalt", "unterbezahlt", "übertariflich",
],
```

Final Database-Structure: Kununu



Analysis
Platform comparison



Biggest deviations:

Interpersonal Relations Pros:

24% vs. 44%

Advancement Cons:

5% vs. 18%

Note. N = 13.649 Kununu Reviews. N = 114.325 Glassdoor Reviews. Chartsshow Herzberg-Factor Prevalences.

Analysis
Platform comparison

Spearman rank correlation of factor prevalences: Kununu vs. Glassdoor

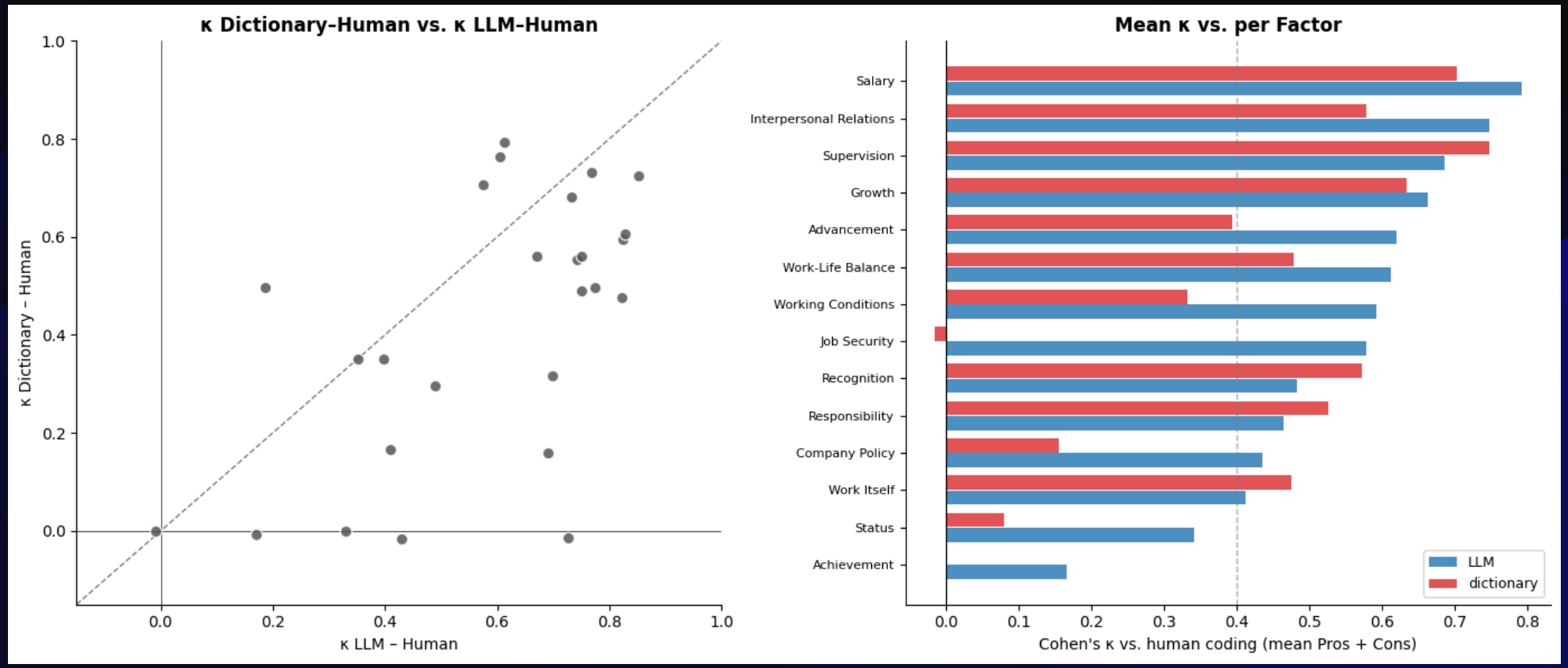
All 28 (14 factors × Pros/Cons)	ρ = 0.804
Pros only (N = 14)	ρ = 0.798
Cons only (N = 14):	ρ = 0.833

Analysis
Agreement of Classification

	Dictionary	LLM	Human
Dictionary	1.000		
LLM	**0.331**	1.000	
Human	**0.404**	**0.602**	1.000

Note. N = 222 classified Reviews. Measured with Cohen's Kappa.

Analysis
Agreement of Classification



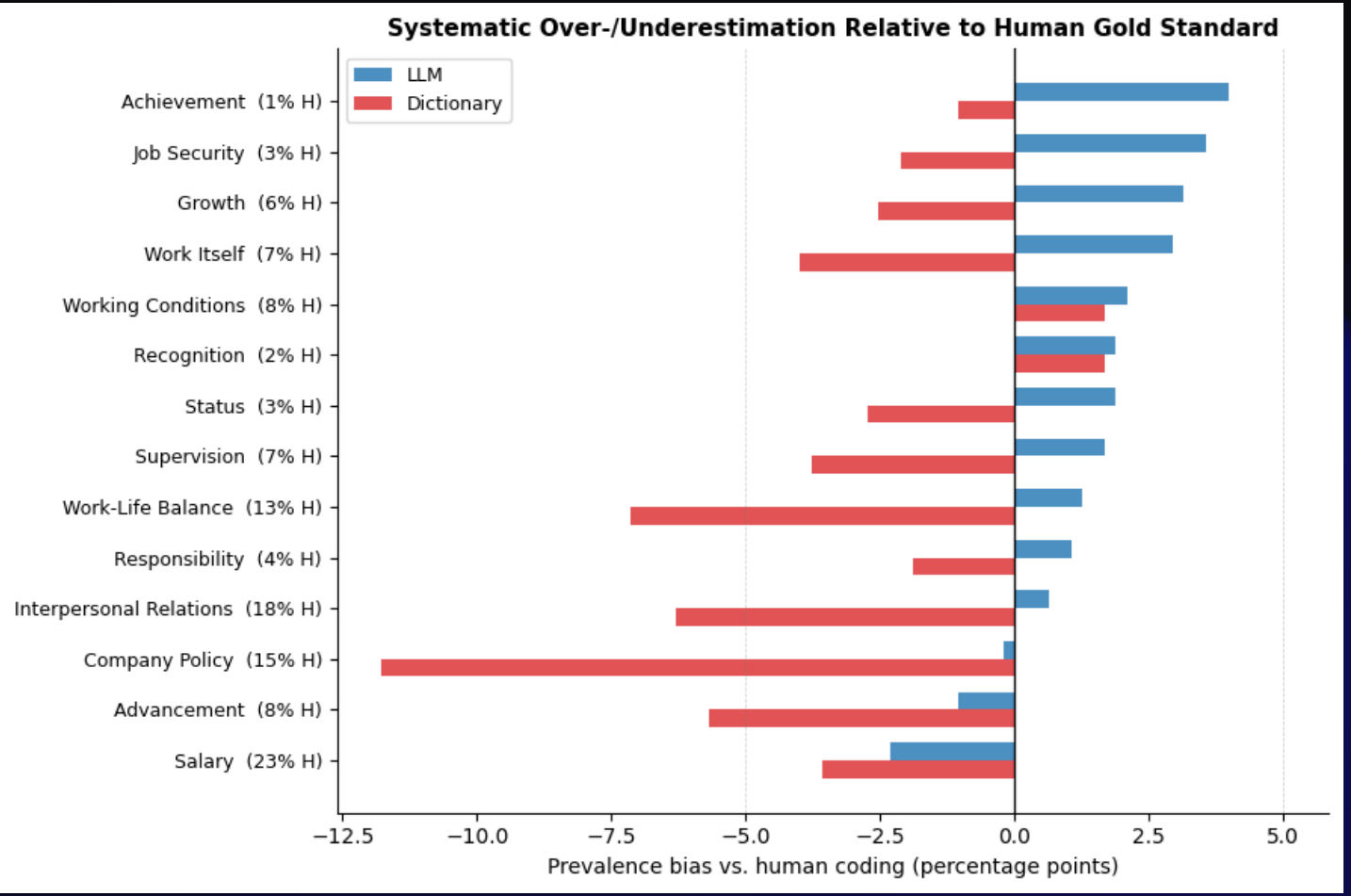
Note. N = 222 classified Reviews. Measured with Cohen's Kappa.

Analysis
Prevalence Bias

Average Bias

LLM: +0.9

Dictionary: -4.79



Discussion

- Agreement between Glassdoor and Kununu is high in terms of factor ranking
- LLM-approach preferred method to classify employee-satisfaction factors
- Would be interesting to assess Reliability of Glassdoor Classification

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Thanks for your attention!

Sources: Literature

Herzberg, F., Mausner, B., & Snyderman, B. (1959). The motivation to work. Wiley.

Sources: Images

Glassdoor: https://upload.wikimedia.org/wikipedia/commons/thumb/f/f0/Glassdoor_Logo_2023.svg/1280px-Glassdoor_Logo_2023.svg.png

Motivation to work: https://m.media-amazon.com/images/I/71vVy9TvKmL._AC_UF894,1000_QL80_.jpg

SQLite: https://upload.wikimedia.org/wikipedia/commons/thumb/3/38/SQLite370.svg/3840px-SQLite370.svg.png

Kununu: https://arbeitgeber-support.kununu.com/hc/theming_assets/01J2V5WPXBSK90K9T85R525BB7